IYYANAR S

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OBJECTIVE:

Biomedical Engineering student passionate about AI in Healthcare, Precision Medicine, and Medical Data Analysis. Interested in developing intelligent diagnostic systems, biomedical signal processing, and predictive healthcare solutions using machine learning and data analytics.

AREA OF INTEREST:

-Medical Imaging (MRI, CT, Ultrasound) and Image Processing Medical Device Design
-**Technical System Design:** Electronics and Mechanical aspects of biomedical equipment
-Digital Health: Biomedical Signal Processing

EDUCATIONAL QUALIFICATION:

Degree\award	College\school	CGPA/%	Year
B.E-BME	PSG College of Technology, Coimbatore	8.0	2023-2027
Class 12	SRI LAKSHMI VIDHYALAYA MATRIC HR SEC SCHOOL, VILLUPURAM	91.1%	2023
Class 10	SRI LAKSHMI VIDHYALAYA MATRIC HR SEC SCHOOL, VILLUPURAM	Pass	2021

INTERNSHIP:

Biomedical Engineer| Medex Technologies Pvt. Ltd., Coimbatore *18 May 2026 –20 June 2026*
Worked hands-on with anaesthesia machine assembly, PCB design using Altium Designer, component selection, soldering, desoldering, and hardware integration. Also gained practical exposure to PCB testing, quality checks, and the overall medical device development process.

Biomedical Engineer Trainee | Popular Nursing Home, Bangalore *8 Dec 2026 – 13 Dec 2026*
Assisted in equipment rounds and learned troubleshooting of biomedical devices. Gained understanding of biomedical equipment in patient care. Observed laparoscopic surgery procedures

PROJECTS:

-Development of a Non-Contact Heart Rate and Respiration Rate for an Emergency Clinical Setting: Designed a contactless vital sign monitoring system using mmWave radar for real-time Heart Rate (HR) and Respiration Rate (RR) estimation through chest motion sensing. Applied signal processing techniques, including phase extraction, filtering, FFT analysis, and HR/RR band separation for accurate physiological monitoring. Developed a real-time healthcare monitoring solution to improve patient comfort, reduce infection risk, and support emergency care applications.

-Thyroid Nodule Detection & Segmentation: Developed an AI-based thyroid ultrasound image segmentation system using U-Net with ResNet34 backbone on the TN3K dataset for accurate thyroid nodule detection and mask prediction. Implemented preprocessing, custom PyTorch data pipelines, Grad-CAM visualisation, and optimised training using Adam optimiser, BCE with LogitsLoss, and Dice Coefficient evaluation. Achieved reliable segmentation performance for medical image analysis using Python, PyTorch, OpenCV, and deep learning techniques.

-Automated Colorectal Polyp Detection & Segmentation – A Deep Learning System for Early Colorectal Cancer Prevention: Developed an automated colorectal polyp detection and segmentation system using a U-Net deep learning architecture on the Kvasir-SEG dataset with Dice + BCE hybrid loss for accurate early-stage colorectal cancer prediction and medical image segmentation.

-Contactless Patient Monitoring System using PIR and Temperature Sensors: Developed a contactless patient monitoring system using Arduino UNO, PIR, and LM35 sensors for real-time body temperature and motion detection with automated buzzer and LED alerts for abnormal patient conditions.

SKILLS:

Programming language: C, Python, Java, and JavaScript (Beginner level), MATLAB.	**Languages Known:** Tamil, English
Tools & Platforms: Arduino, Multisim, PSpice, Adobe Photoshop and Illustrator, Raspberry Pi, MATLAB, Altium Designer	**Workshop**: AI-Powered Embedded Systems, Product Prototyping for Industry Applications
Certifications: Java and JavaScript (Simplilearn), MATLAB onramp (MATLAB), Image processing on-ramp (MATLAB), Introduction to deep learning (GUVI), Medical Image Processing (Coursera), PCB Basic Design Course (Altium Education)	**Soft Skills:** Teamwork, Communication, Organisation, Leadership.

POSITION AND RESPONSIBILITY:

- **Design Team Head**, Biomedical Engineering Association, PSG College Of Technology
- **Design Team Head**, IEEE Students Chapter 12951, PSG College Of Technology
- **Publicity Team Member**, IETE Students' Chapter at PSG Technology
- Placement representative.

EXTRACURRICULAR ACTIVITIES: Photography, Video Editing, Playing Cricket, E-sports, Fitness and Strength Training.